Life and climate, a co-production

The histories of Earth's climate and ecosystems are completely connected, as a smart, whistle-stop tour of the past makes clear, says **Michael Marshall**

Book

The Story of Earth's Climate in 25 Discoveries

Donald R. Prothero

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IT IS is a truism to say that life and climate are intertwined. Living organisms can change Earth's climate by, say, pumping out or absorbing the greenhouse gases that warm the planet. Equally, climate affects life by making conditions too hot, too cold or too dry for some organisms to survive.

This means that the story of life is also the story of climate, as *The Story of Earth's Climate in 25 Discoveries: How scientists found the connections between climate and life* makes clear. Its author, palaeontologist Donald Prothero, takes on the hefty task of recounting how life and climate have shaped each other during Earth's 4.5-billion-year history. Amazingly, he mostly succeeds.

The book is the latest in a series that are essentially extended listicles – with all the good and bad that implies. In his previous books, Prothero told us about dinosaurs in 25 discoveries, life in 25 fossils and Earth in 25 rocks. This time, it is 25 discoveries about past climates and how these shaped life and were shaped by it.

The 25-chapter format keeps things snappy. Each is rarely more than 20 pages long and there are a lot of photographs and illustrations. It is very much a whistle-stop tour.

Prothero tells the story chronologically, beginning with Earth's formation. In fact, he kicks off with the big bang, the expansion of the universe and the likelihood of extraterrestrial life, which – as he acknowledges – is stretching the definition of "climate" just a bit. I found these early chapters frustrating: although they were well-told, I wanted Prothero to get on with it.

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Once the first life took hold on Earth, it started influencing the climate

However, from the third chapter onwards, he gets stuck into the early history of Earth, showing that he has a firm handle on issues such as the origin of the planet's water and the composition of the early atmosphere. He also regularly recounts snippets of science history. Astronomer Carl Sagan makes an early appearance, pointing out that the newly formed sun was fainter than it is now, so the young Earth must have kept itself warm, perhaps with greenhouse gases.

The book rapidly skips over billions of years, with chapters on the rise of atmospheric oxygen, caused by photosynthetic microorganisms, and on the Snowball Earth episodes, when most or all of the planet froze over in truly extraordinary ice ages.

In the second half of the book, Prothero focuses on the past 600 million years, when complex animals and plants appeared and spread all round the world. Each of the "big five" mass extinctions gets its own chapter, as do the switches between greenhouse and icehouse climates on Earth – accompanied by enormous rises and falls of sea level.

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These chapters can get bogged down in palaeontology: in his zeal to describe ecosystems, Prothero sometimes smothers passages in species names. At the risk of being accused of dumbing down, I think these would have benefited from less technical detail and more evocative imagery. I found myself skipping ahead to get to the next climatic shift.

On the positive side, the chapter on the end-Cretaceous extinction is a knockout. Prothero has a big problem with the notion that an asteroid impact was the sole cause of this mass extinction, and also with what he sees as an excessive focus on dinosaurs at the expense of other groups. The whole tone and pace shifts as Prothero moves into fifth gear and explains why it is a bit more complicated than "big rock hits, everything dies".

For someone like me who has been writing about the role of climate in evolution for over a decade, the material in the book was largely familiar and surprises were few and far between. But for the vast majority (including scientists from unrelated specialisms), Prothero's book will be an excellent quick-fire guide to the topic. It is rich with concepts, stories and characters – and there is always another apocalypse just a few pages away. ▮

Michael Marshall wrote *The Genesis Quest: The geniuses and eccentrics on a journey to uncover the origin of life on Earth*. He is based in Devon, UK